

Supporting Information for

Title:

Discovery-based approach to identify multiple factors that affect the spin state of coordination complexes using the Evans NMR method

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1. What is the d electron counts of the following metal ions?(1) Cr^{3+} (2) Co^{2+} **2. Each of the following ions could form a coordination complex with octahedral geometry. Draw d-orbital splitting diagrams of both a) high-spin and b) low-spin octahedral complexes to show orbital occupancies with electrons.**(1) Mn^{3+}

a) High-spin

b) Low-spin

(2) Co^{3+}

a) High-spin

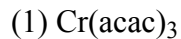
b) Low-spin

2023 Post-lab Quiz

Name:

3. For each of the following complexes, answer questions (a) – (c).

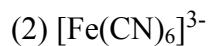
- (a) Determine the oxidation number of the metal.
- (b) Determine the coordination number of the metal (i.e. the number of atoms or ions immediately surrounding the metal in the complex).
- (c) What is the expected geometry of the complex. Select from the following i) – iv)
 - i) Trigonal planar, ii) Tetrahedral, iii) Square planar, iv) Octahedral



(a):

(b):

(c):



(a):

(b):

(c):



(a):

(b):

(c):

2023 Post-lab Quiz

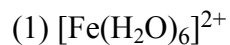
Name:

4. For each of the following complexes, answer questions (a) – (c).

(a) Predict whether this complex is high-spin or low-spin state. Briefly explain your reasoning.

(b) What is the number of unpaired electrons in this complex?

(c) Is this complex paramagnetic or diamagnetic?



(a):

(b):

(c):



(a):

(b):

(c):

5. Answer the following questions.

(1) Which of the following complexes has the largest crystal field splitting energy Δ_o ? Briefly explain your reasoning.

(a): $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, (b): $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$, (c): $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$

(2) Which of the following complexes has the largest crystal field splitting energy Δ_o ? Briefly explain your reasoning.

(a): $\text{Fe}(\text{acac})_3$, (b): $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, (c): $[\text{Fe}(\text{CN})_6]^{3-}$,

(3) Which of the following parameters effect on the magnitude of the crystal field splitting energy Δ_o of coordination complexes? **Select all.**

- (a): type of ligands
- (b): type of metals
- (c): oxidation state of metals
- (d): concentration of coordination complexes in solution

Ans:
